

The Early Prediction of Chronic Diseases Through Diverse Machine Learning Techniques

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Abstract— *Advancements in healthcare analytics offer significant benefits for both healthcare providers and patients. The application of analytics in healthcare plays a crucial role in early disease detection and diagnosis, contributing to enhanced healthcare quality and improved patient outcomes. Machine learning models play a pivotal role in identifying patterns within data, enabling the generation of predictions for disease diagnosis, prognosis, and treatment. With continuous advancements in algorithms and technology, these models have become increasingly effective in providing optimal patient care. This research focuses on utilizing various machine learning algorithms to predict the diagnosis of chronic disease. The study aims to assess and compare the effectiveness of these models by evaluating their prediction accuracy, as well as other performance metrics such as precision, recall, and F1 score. Four different chronic disease datasets considered which includes diabetes, stroke, cardiovascular disease, and Chronic Kidney Disease. Among the investigated models, Random Forest emerged as the most effective, achieving an impressive accuracy rate of 70 to 99% depends on the dataset.*

Keywords—*Cardiovascular Disease, Chronic Kidney Disease, Stroke, Diabetes, Machine Learning, Random Forest.*

I. INTRODUCTION

Artificial Intelligence (AI) is an emerging technology that represent intelligent behavior using computer knowledge, Machine Learning (ML) techniques as well as insignificant human involvement. AI is gradually incorporated into scientific innovations to enhance and hasten investigations, assisting scientists to create hypotheses, design experiments, accumulate and interpret large datasets, and also comprehend the results which might have not been possible only through traditional scientific methods [1].

In the healthcare sector, AI discovers diverse applications to build state of the art machines that can be able to perceive diseases and detect cancer cells efficiently. By the use of lab and other medical data, Artificial Intelligence provides assistance to analyze chronic conditions and ensure early diagnosis of such chronic disease.

The noncontiguous diseases which includes diabetes, numerous cancers, Cardiovascular diseases (CVDs), chronic respiratory illness, and so on which are the major causes as estimated 60% of the global deaths. It is observed that 17.7 million deaths caused by CVDs like ischemic heart disease and cerebrovascular disease such as stroke [2]. The five adaptable risk factors includes Systolic blood pressure, BMI, Non-HDL

cholesterol, smoking and diabetes are associated with Cardiovascular disease and may be a cause of death.

Diabetes is one of the most common global diseases as 37.3 million Americans affected by diabetes mellitus in the year 2019, it was approximately 11.3% of the whole country's population. The diabetes-related deaths proportion had been estimated as in between 11.5% to 12% in the US, where the rate of obese people were significantly higher at 19.4% [3]. Although, it is expected that 1.3 billion individuals will get influence of diabetes by 2050 as it is an ever increasing global health issue [21].

Many researchers adopt the technique of data generation through simulation, sufficient amount of healthcare data could be achieve with an insufficient amount of sample. Song Yu et al. [22] is one of them who generated healthcare data through simulation which then validated by SVM model. Ahmed, Z., Naz, S. et. al [23] has also generated data for vehicular movements through simulation. Vehicular mobility patterns has been generated under different scenarios using VanetMobiSim (a realistic vehicular movement trace generator use for telecommunication networks for large-scale simulation) to emulate realistic vehicular environment. Conversely, Ubaida Fatima et. al (2003) investigated complex-real life healthcare datasets and catered in various situations [24].

This study concentrates on employing diverse machine learning algorithms for forecasting chronic disease diagnoses. Its objective is to gauge and contrast the efficacy of these models through the assessment of their predictive accuracy along with other performance metrics like precision, recall, and F1 score. Four distinct chronic disease datasets will be analyze, encompassing diabetes, stroke, cardiovascular disease, and Chronic Kidney Disease.

II. RELATED WORK

In 2023, Yashvanth, R. et al studied and underlined the limitations and potential of Machine Learning models in the field of healthcare in [4], And take first step to initiate future research in this domain of early prediction and diagnosis of diabetes which is a silent killer. In this research work twelve algorithms has been used. Results taken from this research shows that the maximum mean accuracy score is obtained by the commonly used ML algorithm called Gradient Boosting algorithm. Another analyses has been performed by Sailasya,

Gangavarapu et al. in 2021. In this study [5], the performance of numerous machine learning algorithms by taken multiple physiological factors to predict the possible chances of stroke in brain has been evaluated. From all the considered machine learning algorithms, Naïve Bayes Classifier out performs by proving the 82% accuracy and also performs better among all other remaining essential metrics used in this study. The study in [6] in 2023 explored various machine learning techniques to facilitate early CKD diagnosis. In this research Islam, Md Ariful et al. integrated advanced predictive modeling to improve accuracy. Sanmarchi, Francesco, et al. in 2023 present a comprehensive review in [7], The objective was to evaluate the utilization of AI, incorporating machine learning (ML) methodologies, in predicting, diagnosing, and managing chronic kidney disease (CKD).

A research has been done by Sk, Khader Basha et al. in 2023, They emphases in this study on predicting and classifying coronary heart disease (CHD) through Hybrid Machine Learning methods. The approach contains the combination of Decision Tree along with Ada Boosting algorithms as a hybrid ML algorithm for CHD prediction. [8].

Hypertension, a widespread global health concern, is linked to various risk factors. The significance of early diagnosis of such a silent disease lies in preventing associated health complications. A predictive analysis of the Qatar biobank has been discussed in [9] by AlKaabi, Latifa A. et al. in 2020. The development and assessment of the predictive models is the main objective of this research that can be able to detect individuals at a heightened risk of being a hypertension patient. Fang, Min et al. in [10] proposed a hybrid model in his research in the year 2023. This research tackle the task of predicting the risk of hypertension over the next five years. A hybrid model that combines KNN and LightGBM has been suggested that is used to forecast the likelihood of hypertension for an individual based on features such as age and blood indicators. In reference [11], the prediction of strokes was conducted on the Cardiovascular Health Study (CHS) dataset employing five machine learning techniques.

III. DESCRIPTION OF DATASETS

A. Dataset for Diabetes

The healthcare dataset for diabetes is taken from world largest data community named as Kaggle.com [17]. This considered dataset of diabetes has 70692 rows and 18 feature columns which includes 'Age', 'Sex', 'BMI', 'PhysActivity', 'Diabetes' and etc. The 'Age' column is categorized in 13 levels of age categories. The main attribute 'Diabetes' as the resultant column contains the values either '1' or '0'. The value '0' means that there is no risk of diabetes detected and on the other hand the output '1' designates the patient suffering with possible risk of diabetes. There are 35364 diabetics patients with the output value '1' and same number of patients has no diabetes.

B. Dataset for Cardiovascular Disease

The Cardiovascular Disease dataset is also grab from kaggle [18]. This set of data contains 13 columns and 70000 rows. The parameters of this data includes 'age', 'height', 'ap_hi', 'ap_lo', 'cardio' etc. Since the dataset is already balanced so no need to balance the considered dataset but data cleaning has performed because there are some negative values appeared in 'ap_hi' and 'ap_lo' column.

C. Dataset for Stroke

The dataset for stroke is comprise of 11 attributes and 40910 entries and this data is also grab from kaggle.com [19]. The output variable 'stroke' of the considered dataset holds the values as either '0' or '1' in which the value '1' directs a probable threat of stroke while the value '0' specifies that no risk of stroke detected as output. No data preprocessing is required because this stroke dataset is already balanced as it contains 20460 rows which has the value '1' and 20450 rows which has the value '0'. The dataset contains feature columns as 'age', 'hypertension', 'heart_disease', 'smoking_status', and etc.

D. Dataset for Chronic Kidney Disease

The dataset for Chronic Kidney Disease is comprise of 25 attributes and 400 entries and this data is also chosen from kaggle.com [20] for further validation of model. The output variable 'Chronic Kidney Disease' of the considered dataset holds the values as either '0' or '1' in which the value '1' directs a probable occurrence of kidney disease while the value '0' specifies that no risk of Kidney Disease detected as output. No data preprocessing is required because this dataset is already balanced as it contains 250 rows which has the value '1' and 150 rows which has the value '0' means that no risk of kidney Disease has been detected in overall 400 patients. The dataset contains feature columns as 'age', 'Blood Urea', 'Sodium', 'Potassium', and etc.

IV. METHODOLOGY

A. Techniques used for Machine Learning Analysis

The considered techniques for Machine Learning analysis on various datasets are discussed.

1. Decision Tree Classifier

The Decision Tree classification algorithm is one that is mainly designed to address both regression and classification issues [12]. This classifier is categorized in supervised learning, where the output variables possessed from their corresponding input variables. It follows a tree-like structure, with data continuously splitting based on a specific parameter. The Decision Tree comprises of two components: the Decision Node, where the data splits, and the Leaf Node, which provides the final outcome.

2. Random Forest Classifier

The Random Forest Classifier consist of various independent decision trees where each individual tree trained on a random subset of data. During the training phase all these trees have created, and the outputs are derived from each separable decision tree. At the final stage of prediction of this algorithm, a process occurs which is referred as "voting". In this method, each individual decision tree able to casts a vote for a class which is basically known as an output class, such as 'stroke' or 'no stroke.' The selection of the class through Random Forest with the highest number of votes as the eventual prediction [13]. It is theoretically possible for a Random Forest algorithm to achieve 100% test accuracy on certain datasets, it's exceedingly rare in practice, especially with real-world data. Here's a reference to a research article discussing the

limitations and challenges of achieving perfect accuracy with Random Forests [14].

3. Logistic Regression

Logistic Regression is another supervised learning algorithm that is designed to deal regression related problems, employed to predict the output variable’s probability [15]. This algorithm is particularly suitable when the resultant variable assumes 0 and 1 as the binary values. Given that the attribute which contain output in the considered dataset retains just two potential values, The Logistic Regression is the best method to pick.

4. K-Nearest Neighbor Classifier

K-Nearest Neighbors (KNN) is another algorithm employed for classification. This supervised learning technique, as described by KNN [16], is normally considered as a "lazy" algorithm, meaning it is not able to train immediately upon receiving the dataset. Instead of that, it has the capability to store the information and, during classification process, KNN performs on it.

B. Evaluation Metrics

To assess the performance and effectiveness of a machine learning model, Evaluation metrics are used. These are the quantitative measures that provides the intuitions to check the performance of the model that is the model perform well or not, also give assistance in comparing distinct models or algorithms. It is importunate to assess the predictive ability, overall quality and generalization capacity of a machine learning model when evaluating it. In this regard, Evaluation metrics deliver objective criteria to estimate these features. The selection criteria of the evaluation metrics depends on the type of model and the implementation plan of the model. The four evaluating metrics will choose in this article to evaluate the models’ performance and convincingness. These metrics are Accuracy, Precision, Recall, and F1 Score

V. RESULTS AND INTERPRETATION

1. Cardiovascular Disease

On Cardiovascular disease dataset, we have applied four different machine learning algorithms which includes Decision Tree, Logistic Regression, Random forest classifier, and K- Nearest Neighbor Classifier. From all these four algorithms random forest gave best results as outperform among all with 71.89% accuracy followed by Logistic Regression gave 70.08% accuracy while K-Nearest Neighbor gave the worst output with 54.46% that is the lowest accuracy as shown in Table 1 achieved in all four algorithms. Performance evaluation with other metrics along with accuracy are also shown in Table 1 and in Figure 1, 2, 3 and 4

Table 1: Cardiovascular Disease dataset

Model Used	Accuracy	F1 Score	Precision	Recall
Decision Tree	0.6313	0.6313	0.6313	0.6313
Logistic Regression	0.7008	0.7001	0.7023	0.7008
K-Nearest Neighbor	0.5446	0.5446	0.5446	0.5446
Random Forest	0.7189	0.7185	0.7198	0.7189

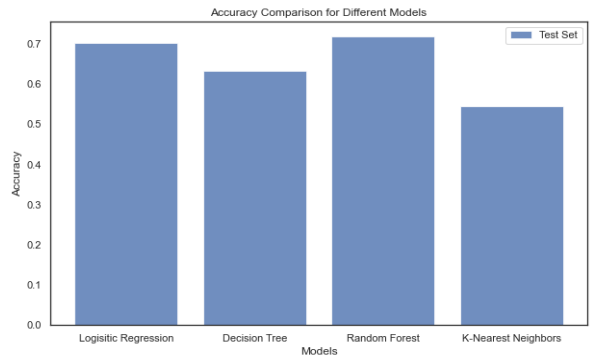


Figure 1: Visualization of Accuracy comparison for Different models on Cardiovascular Disease dataset

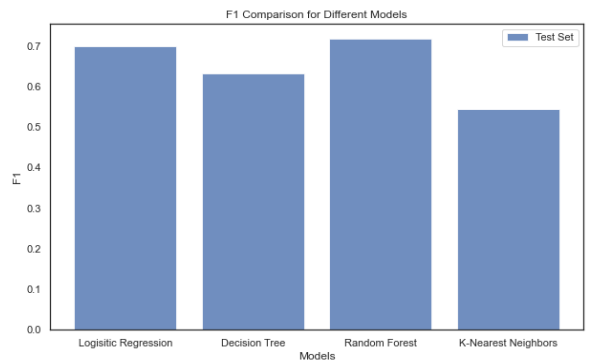


Figure 2: Visualization of F1 Score comparison for Different models on Cardiovascular Disease dataset

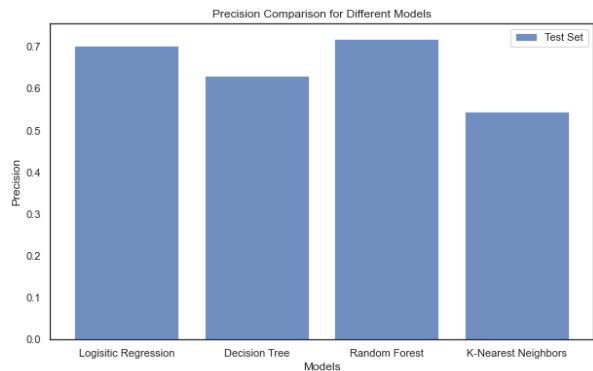


Figure 3: Visualization of Precision comparison for Different models on Cardiovascular Disease dataset

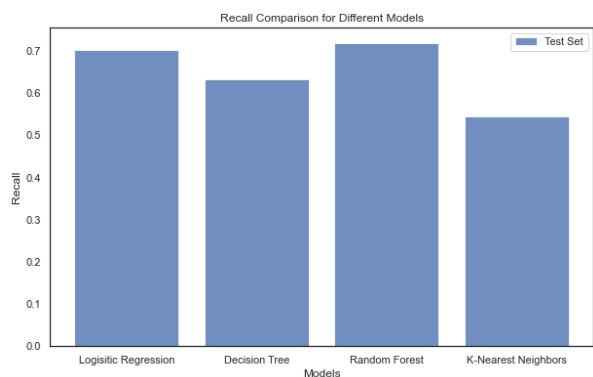


Figure 4: Visualization of Recall comparison for Different models on Cardiovascular Disease dataset

2. Diabetics

The outcomes achieved in term of mean accuracy from this research revealed in Table 5 that the classification algorithms employed demonstrated moderate performance on the dataset

of diabetes, presenting mean accuracy scores in the range from 65% to 75%. From the all four considered algorithms Logistic Regression exhibited the highest mean accuracy score at 74.64%, After that the Random Forest Classifier (72.95%) then K-Nearest Neighbor Classifier (69.48%) and the lowest accuracy achieved on this dataset of diabetes by Decision Tree Algorithm of 65.80%. The Table 2 illustrates the mean accuracy and other performance evaluation metrics obtained from all four machine learning algorithms used. Figure 5, 6, 7, and 8 represents the accuracy, F1 score, precision, and recall score of this model respectively.

Table 2: Diabetes Dataset

Model Used	Accuracy	F1 Score	Precision	Recall
Decision Tree	0.6580	0.6578	0.6581	0.6580
Logistic Regression	0.7464	0.7463	0.7470	0.7464
K-Nearest Neighbor	0.6948	0.6947	0.6952	0.6948
Random Forest	0.7295	0.7292	0.7309	0.7295

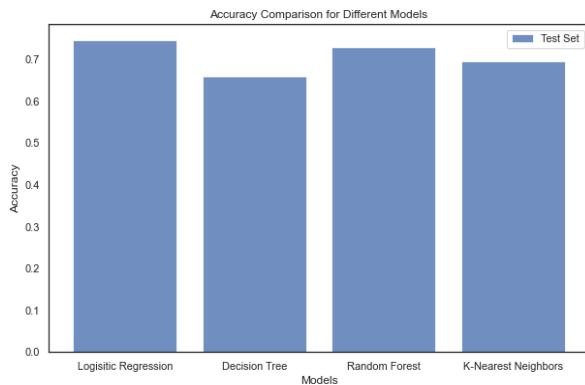


Figure 5: Visualization of Accuracy comparison for Different models on Diabetes dataset

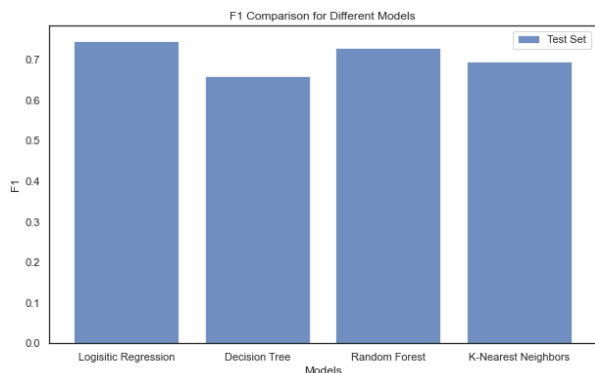


Figure 6: Visualization of F1 Score comparison for Different models on Diabetes dataset

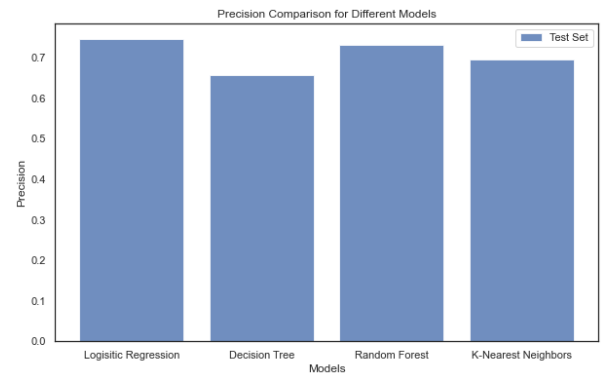


Figure 7: Visualization of Precision comparison for Different models on Diabetes dataset

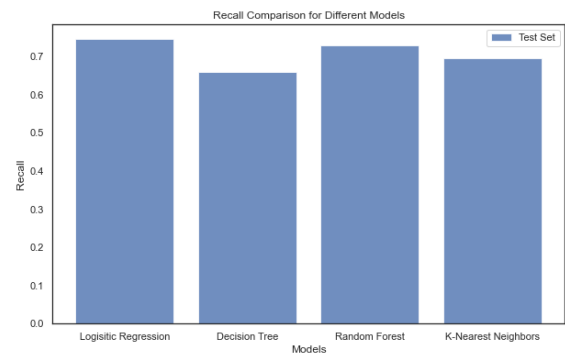


Figure 8: Visualization of Recall score comparison for Different models on Diabetes dataset

3. Stroke

In the analysis of the Stroke dataset, we utilized four different machine learning algorithms: Decision Tree, Logistic Regression, Random Forest Classifier, and K-Nearest Neighbor Classifier. Notably, Decision Tree and Random Forest demonstrated similar outcomes, surpassing the rest with a 99% accuracy rate. On the contrary, Logistic Regression produced the least favorable results, recording the lowest accuracy among the all four algorithms at 68.20%. The table 6 presents the accuracy along with other metrics of each considered machine learning algorithm while Figure 9, 10, 11, and 12 depicts the performance of all four algorithms in detail.

Table 3: Stroke Dataset

Model Used	Accuracy	F1 Score	Precision	Recall
Decision Tree	0.9994	0.9994	0.9994	0.9994
Logistic Regression	0.6820	0.6805	0.6849	0.6820
K-Nearest Neighbor	0.8511	0.8488	0.8750	0.8511
Random Forest	0.9968	0.9968	0.9968	0.9968

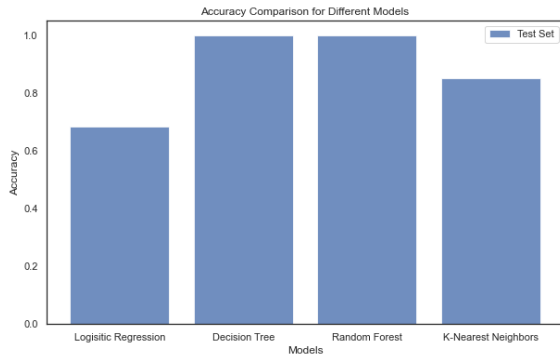


Figure 9: Visualization of Accuracy comparison for Different models on Stroke dataset

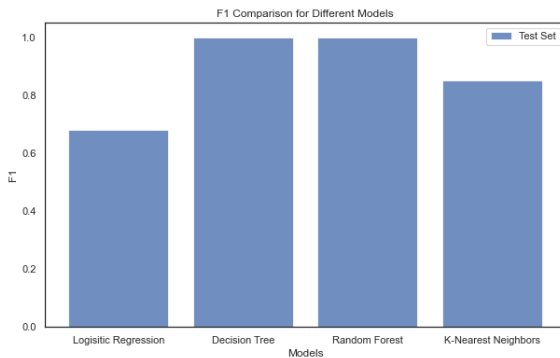


Figure 10: Visualization of F1 Score comparison for Different models on Stroke dataset

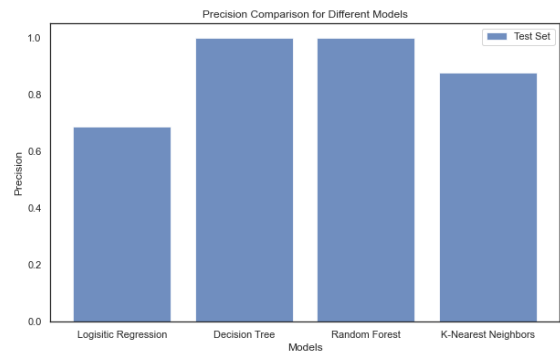


Figure 11: Visualization of Precision comparison for Different models on Stroke dataset

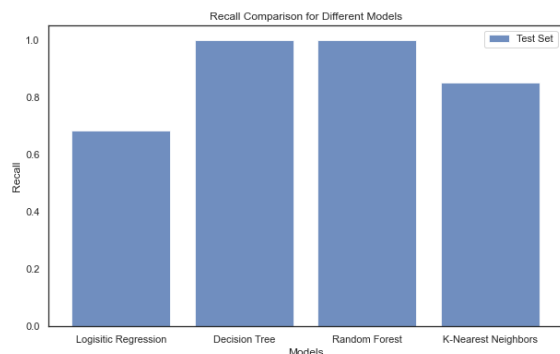


Figure 12: Visualization of Recall score comparison for Different models on Stroke dataset

4. Chronic Kidney Disease

The results obtained regarding mean accuracy in this study, as presented in Table 7, indicate that the utilized classification algorithms demonstrated moderate performance when applied to the Chronic Kidney Disease dataset, with mean accuracy scores ranging from 66% to 100%. Among the four algorithms

considered, Random Forest Classifier exhibited the highest mean accuracy score at 100%, followed by the decision Tree Classifier at 99.38%, and then the Logistic Regression at 98.12%. Table 4 displays the mean accuracy obtained from all four machine learning algorithms used. Additionally, Figures 13, 14, 15, and 16 represent the accuracy, F1 score, precision, and recall scores of this model, respectively.

Table 4: Chronic Kidney Disease Dataset

Model Used	Accuracy	F1 Score	Precision	Recall
Decision Tree	0.9938	0.9938	0.9939	0.9938
Logistic Regression	0.9812	0.9813	0.9814	0.9812
K-Nearest Neighbor	0.6625	0.6681	0.7207	0.6625
Random Forest	1.0000	1.0000	1.0000	1.0000

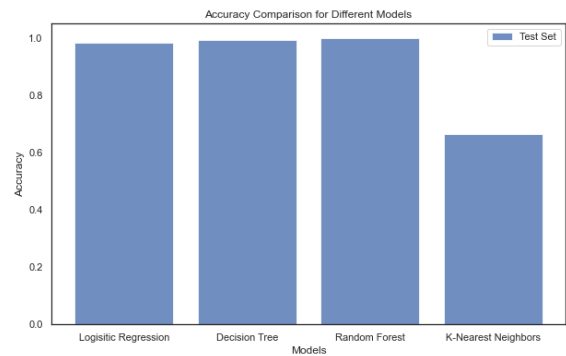


Figure 13: Visualization of Accuracy comparison for Different models on Chronic Kidney Disease dataset

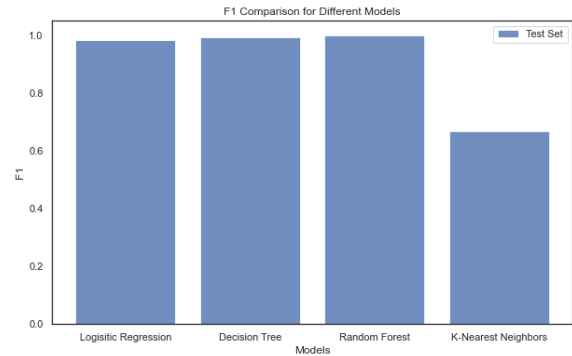


Figure 14: Visualization of F1 Score comparison for Different models on Chronic Kidney Disease dataset

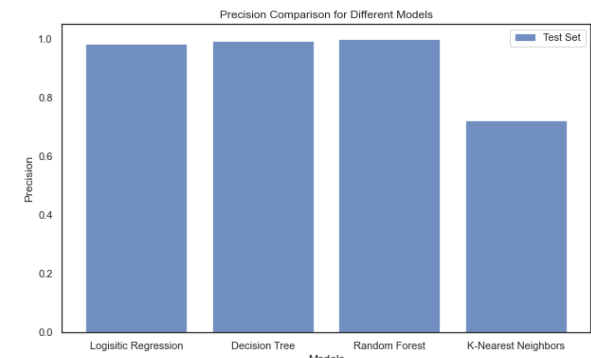


Figure 15: Visualization of Precision comparison for Different models on Chronic Kidney Disease dataset

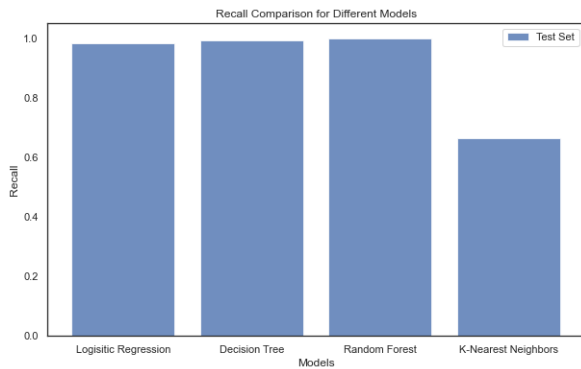


Figure 16: Visualization of Recall score comparison for Different models on Chronic Kidney Disease dataset

VI. CONCLUSION AND FUTURE DIRECTIONS

Chronic disease like Cancer, Hypertension, Chronic Kidney Disease, Cardiovascular Disease, Chronic Lungs Disease, stroke and arthritis etc. become the most common cause of disability and even death in all over the world. It is mainly the condition that a person suffering from the last one or more years and also required medical attention or need to manage daily activities. This study deepen our consideration of developing some machine learning models for the diagnoses of such chronic disease at their earliest which includes stroke, diabetes, and cardiovascular diseases, providing valuable intuitions into the competences of considered algorithms within the healthcare domain. For the Cardiovascular disease dataset, both random forest and logistic regression algorithms yielded comparable results, outperforming all others with a 70% to 72% accuracy rate. Concerning the diabetes dataset, the Logistic Regression algorithm demonstrated the mean accuracy score of 74.64% which is the highest accuracy score obtained by the considered dataset of diabetes, the second mean accuracy found by the Random Forest Classifier at 72.95%. In the examination of the Stroke dataset, both Decision Tree and Random Forest exhibited similar outcomes, surpassing the other algorithms with an accuracy rate of 99%. Regarding the Chronic Kidney Disease dataset, the Random Forest algorithm displayed the highest mean accuracy score 100%, which stands as the top accuracy score among the considered algorithms for this dataset.

Among the all four models explored, Random Forest proved to be the most effective, demonstrating an impressive accuracy rate ranging from 70% to 100%, depending on the dataset. To validate the outcomes of our study and to measure the algorithms' potential in an extensive and more diverse population, additional investigations are needed. In summary, this research underscores the significance of employing numerous machine learning algorithms for discovering the timely diagnoses of chronic disease like diabetes, cardiovascular disease, Chronic Kidney Disease and stroke laying the foundation for future studies in this field of healthcare.

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